PROJECT REPORT

Blinkit Sales Analysis Dashboard Using Power BI

Submitted By:

Lokesh Ramlal Chaudhari

Course:

Bachelor of Computer Applications (BCA)

Department:

Computer Applications

College:

G.H. Raisoni college of Engineering and Management, JALGAON.

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Introduction & Problem Statement

Introduction

In the modern retail and quick-commerce industry, companies generate large volumes of sales data from product transactions, outlet operations, and customer ratings. Analyzing this data helps businesses understand customer behavior, improve inventory management, and optimize sales strategies.

This project focuses on analyzing a Blinkit-style retail sales dataset using **Power BI** to extract meaningful insights about product performance, outlet distribution, and sales trends. Power BI is a powerful business intelligence tool that allows users to transform raw data into interactive visual dashboards.

By creating an interactive dashboard, stakeholders can easily explore sales performance, compare product categories, and evaluate outlet efficiency across different locations.

Problem Statement

Quick-commerce companies operate across multiple outlets and cities, generating large volumes of sales data. Without proper analysis, it becomes difficult to understand which products perform well, how outlet characteristics affect sales, and how customer preferences vary across regions.

The objective of this project is to analyze sales data and develop an interactive dashboard using Power BI to visualize key performance indicators and identify patterns related to product demand, outlet performance, and customer satisfaction.

Methodology & Dashboard Screenshot

Methodology

The project was implemented using **Power BI** as the primary data analysis and visualization tool. The dataset used in this project contained approximately **8,523 records and 12 columns**, including information about item type, outlet size, outlet location, sales value, and customer ratings.

Data Cleaning

The dataset required preprocessing before analysis. Inconsistent values in the **Item Fat Content** column were standardized. For example:

- "LF" and "low fat" were converted to **Low Fat**
- "R" and "reg" were converted to **Regular**

The cleaned dataset was then loaded into Power BI for further transformation and analysis.

Data Modeling and Visualization

Power BI features such as **data modeling, calculated measures, and interactive visuals** were used to build the dashboard.

Key performance indicators were calculated including:

- Total Sales
- Average Sales
- Total Items Sold
- Average Customer Rating

Various charts and visualizations were created to analyze sales trends across product categories, outlet sizes, outlet locations, and establishment years.

Dashboard Screenshot:

Results, Insights & Conclusion

Results and Insights

The analysis provided several important insights into sales performance and product demand.

The total sales recorded in the dataset were approximately $1.20 million, with an average sales value of $141 per item. The dataset contained 8,523 items, and the average customer rating was 4.0, indicating overall positive customer feedback.

From the analysis, snack foods and fruits emerged as the top-performing product categories. Medium-sized outlets generated the highest sales compared to small and large outlets. Additionally, the data showed that low-fat products contributed significantly to total sales, indicating growing customer preference for healthier food options.

The sales distribution also highlighted strong demand in Tier 2 cities, suggesting rapid growth of quick-commerce services outside major metropolitan areas.

Conclusion

This project demonstrates how Power BI can be used to transform raw sales data into meaningful business insights through interactive dashboards.

The Blinkit Sales Analysis Dashboard enables users to analyze key metrics, explore sales patterns, and understand customer preferences effectively. Such insights can help businesses improve decision-making, optimize product distribution, and enhance overall operational efficiency.